

Question

The following compound

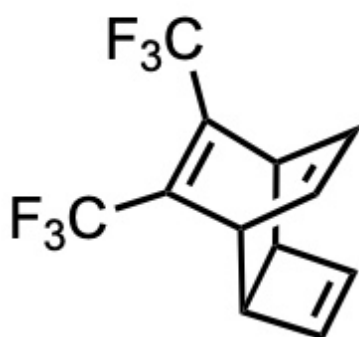


Image is a molecular formula, SMILES code is FC(C1=C(C(F)(F)F)C2C(C=C3)C3C1C=C2)(F)F

After olefin metathesis, polymer **A** is obtained (only one double bond in each monomer participates in olefin metathesis). **A** spontaneously decomposes into a stable small molecule **B** and a well-known polymer **C** at room temperature (the polymer chain does not break). Select the correct option:

- A.** All other options are incorrect
- B.** **C** does not exhibit cis-trans isomerism
- C.** **A** Spontaneous decomposition is entropically unfavorable, but because it releases stable products, it has a large enthalpic driving force, and therefore can proceed forward.
- D.** The molecular weight of small molecule **B** is less than 120.

E. Treating **C** with iodine monochloride can enhance conductivity.

F. **C**'s monomer has three rotational modes.

Answer

Correct Answer: **E**

Detailed Explanation

Observing the molecular structure, it can be seen that the double bond within the four-membered ring in the molecule is very unstable and should be the reaction site for olefin metathesis. Therefore, the monomer structure of polymer **A** is $[R]=CC1C2C(C(F)(F)F)=C(C(F)(F)F)C(C1C=[R])C=C2$, where R is the connection site of the polymer monomer.

CHECKPOINT

1 PTS

The monomer structure of polymer **A** is $[R]=CC1C2C(C(F)(F)F)=C(C(F)(F)F)C(C1C=[R])C=C2$, where R is the connection site of the polymer monomer

CHECKPOINT

1 PTS

The double bond participating in the reaction in olefin metathesis is the double bond within the four-membered ring

Subsequently, it was observed that a reverse 4+2 reaction could occur, removing a molecule with a benzene ring structure and obtaining polyacetylene. Therefore, **C** is polyacetylene, and the structure of **B** is $FC(C1=CC=CC=C1C(F)(F)F)(F)F$.

This process generates stable products, reducing tension while producing more molecules, so the reaction entropy is favorable. According to the molecular formula and reaction process, **C** and **D** are excluded.

CHECKPOINT

1 PTS

Reverse 4+2 reaction occurs, resulting in **C** being polyacetylene

CHECKPOINT

1 PTS

The structure of **B** is FC(C1=CC=CC=C1C(F)(F)F)(F)F

CHECKPOINT

1 PTS

The number of reaction molecules increases, resulting in an increase in entropy

The monomer of polyacetylene is acetylene, which is a linear molecule and has only two rotational modes, excluding *F*.

CHECKPOINT

1 PTS

Acetylene is a linear molecule and does not have three rotational modes

Polyacetylene has many double bonds and should have cis-trans isomers, excluding *B*.

CHECKPOINT

1 PTS

Polyacetylene has cis-trans isomers

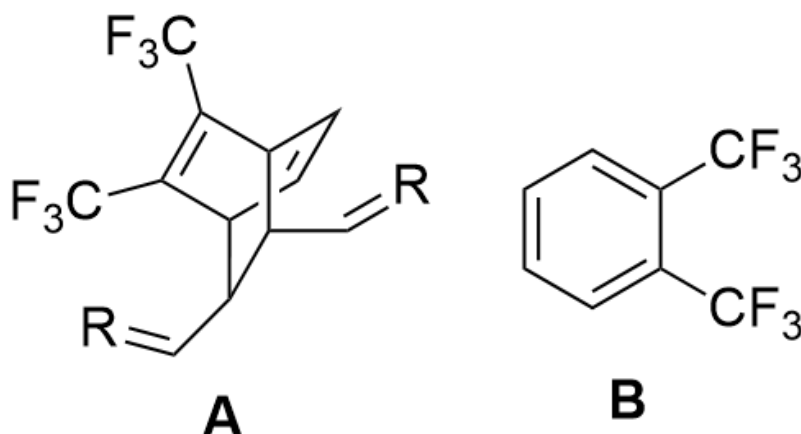
Treating polyacetylene with iodine can result in electron-deficient holes, enhancing conductivity. Therefore, option *E* is correct, and option *A* is incorrect.

CHECKPOINT

1 PTS

Treating polyacetylene with iodine can result in electron-deficient holes, enhancing conductivity

Therefore, the answer is option E.



The monomer structure of polymer **A** is $[\text{R}]=\text{CC1C2C}(\text{C}(\text{F})(\text{F})\text{F})=\text{C}(\text{C}(\text{F})(\text{F})\text{F})\text{C}(\text{C1C}=[\text{R}])\text{C}=\text{C2}$, where R is the connection site of the polymer monomer; the structure of **B** is $\text{FC}(\text{C1}=\text{CC}=\text{CC}=\text{C1C}(\text{F})(\text{F})\text{F})(\text{F})\text{F}$.